

Model-based Survey Weighting Using Hierarchical Gaussian Processes for Structured Populations

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Abstract. Survey weighting is a central tool for producing population-representative estimates in the presence of unequal sampling probabilities, nonresponse, and undercoverage. While classical design-based weights rely on known inclusion probabilities, modern surveys increasingly require model-based approaches to handle high-dimensional poststratification spaces and sparse cell counts. Building on prior work connecting logistic regression to equivalent survey weights, this paper develops a flexible Bayesian framework for survey weighting using hierarchical Gaussian process (HGP) regression. The proposed model decomposes the latent outcome-generating process into structured group-level effects and a smooth kernel-based component, allowing information to be shared across related poststratification cells while accommodating nonlinear and interaction effects. We derive closed-form expressions for posterior predictive means and covariances under monotonic link functions, and show how structured covariance representations yield substantial computational savings relative to standard Gaussian process models. A key theoretical contribution is a uniform convergence result for the posterior predictive mean under mild regularity conditions. Leveraging this result, we establish a formal connection between model-based poststratification and survey weighting by introducing *local equivalent unit weights*, defined via a first-order Taylor expansion of the poststratified estimator around the fitted latent mean surface. These weights generalize classical equivalent-weight representations for generalized linear models to the nonparametric Gaussian process setting, and admit simple closed forms for identity, logit, and probit links. Simulation studies demonstrate that the proposed HGP-based weights reduce bias and variance relative to logistic regression and kernel-based alternatives, particularly in sparse and highly structured populations. The results provide a principled and scalable approach to constructing survey weights that integrates modern Bayesian nonparametric modeling with classical design-based intuition.

1 Introduction

Survey weighting is a fundamental tool for correcting selection bias and achieving population-representative estimates, especially when data suffer from unequal sampling probabilities, nonresponse, or undercoverage. In our previous work, we explored a model-based framework for constructing base survey weights using logistic regression (Lee and Gelman, 2026). That approach offered a structured alternative to traditional design-based weights, showing how predictive modeling could align with classical estimators like the Hájek estimator when appropriately normalized. However, logistic regression—while useful—assumes a linear structure on the log-odds scale and does not easily accommodate complex interactions or smooth variation across structured populations ().

In this sequel, we introduce a more flexible model-based approach to survey weight construction using Hierarchical Gaussian Processes (HGPs). HGPs offer a principled way to model complex, nonlinear relationships across both continuous and discrete covariates, while borrowing strength across groups through hierarchical structure. This is particularly valuable in survey settings where

small subgroup sizes or sparsely observed cells limit the stability of traditional poststratification or regression-based weighting methods (Little, 1993; Gelman and Little, 1997; Park et al, 2004; Si et al, 2020).

A key computational innovation in our method is the use of structured covariance matrices that exploit the known grouping of the population. Rather than fitting a full Gaussian process across individual-level covariates—which can be computationally expensive due to the $\mathcal{O}(n^3)$ scaling (Rasmussen and Williams, 2006)—we construct a covariance matrix over groups (e.g., poststratification cells from weighting variables) (Banerjee et al, 2008; Gelman et al; 2013, Si et al, 2015, Makela et al, 2018; Roustant et al, 2020). This reduces the dimensionality of the GP to the number of unique groups rather than individuals, yielding substantial computational savings while preserving flexibility in the modeled inclusion probabilities. In practice, this makes HGP-based weighting feasible even for large surveys with thousands of respondents and of poststratification cells.

We develop this approach within a fully Bayesian framework and compare the resulting population estimates to those from alternative models, and the resulting weights to both our previous model-based methods and classical design-based approaches (Little, 1991; Deville and Särndal, 1992; Gelman, 2007; Si et al, 2015). Our simulations demonstrate that the HGP-based method not only reduces bias in estimated outcomes and variance in estimated weights but also enhances robustness in the presence of data sparsity, providing more stable inference without requiring external raking or trimming. While traditional calibration and trimming procedures can mitigate extreme weights, they often introduce additional tuning choices and instability in sparse settings (Little and Vartivarian, 2003; Haziza and Beaumont, 2017; Chen et al, 2020). Additionally, because binary-outcome models such as logistic regression are particularly sensitive to sparsity and separation in finely stratified cells (Albert and Anderson, 1984; Chen et al, 2010; Kosmidis and Firth, 2021), our framework is designed to accommodate a broad range of outcome types, ensuring wider applicability across survey settings.

This paper contributes a flexible and scalable model-based framework for survey weighting, with a focus on structured populations. Our results support the use of HGPs for smoothing across cells and correcting for complex inclusion mechanisms—expanding the practical toolbox available for researchers seeking principled and efficient weighting strategies.

2 Hierarchical Gaussian Process Regression

Our goal is to estimate the population mean of an outcome variable y across structured subgroups, given a sample that sparsely covers the full poststratification space. Let the population be divided into H hierarchical groups (e.g., demographic strata for which population-level information is available) and K kernel groups (e.g., the full observed set of poststratification cells spanning all demographic combinations). In typical survey settings, we observe only a small subset of the full set of poststratification cells (e.g., 1,100 observed cells out of 4,480 total), which leads to instability in direct cell-level estimation (Lee and Gelman, 2024).

To overcome this sparsity, we propose a hierarchical Gaussian process (HGP) model that enables information sharing across related groups (Si et al, 2015; Roustant, 2020; Li et al, 2025). Specifically, we assume that the latent outcome-generating process is composed of additive components: a structured group-level effect μ_h and a smooth kernel-based effect η_k modeled by a Gaussian process.

Formally, for each observation $i = 1, \dots, n$ in the sample, we assume the following model:

$$\mathbb{E}[Y \mid X] = g^{-1}(f(X)) + \mathbb{E}[\varepsilon], \quad \varepsilon \sim \mathcal{N}(0, \sigma^2),$$

where $Y \in \mathbb{R}^n$, $X \in \mathbb{R}^{n \times p}$, $g(\cdot)$ is a monotonic link function, and the latent function is defined component-wise as

$$f_i = \mu_{h[i]} + \eta_{k[i]}, \quad \text{for } i = 1, \dots, n.$$

where $\mu_{h[i]}$ is the fixed effect for group $h[i] \in \{1, \dots, H\}$ and $\eta_{k[i]}$ is the kernel-based effect for group $k[i] \in \{1, \dots, K\}$. For continuous outcomes, we set $\sigma^2 > 0$. For binary outcomes using a non-identity link (e.g., probit or logit), we set $\sigma^2 = 0$ to reflect the deterministic link from the latent scale to the probability scale.

We place the following priors on the latent components:

$$\mu \sim \mathcal{N}(\boldsymbol{\mu}_{\text{prior}}, \Sigma_{\mu}), \quad \eta \sim \mathcal{GP}(0, K_{\eta}),$$

where $\boldsymbol{\mu}_{\text{prior}} \in \mathbb{R}^H$ encodes known population-level means, and K_{η} is the Radial Basis Function

(RBF) kernel defined over K structured groups. The squared exponential kernel is computed via:

$$[K_\eta]_{ij} = \alpha^2 \exp \left(-\frac{\|X_i - X_j\|^2}{2\rho^2} \right),$$

where $K_\eta : \mathbb{R}^P \times \mathbb{R}^P \rightarrow \mathbb{R}$ given that X_i and X_j are P -dimensional covariate vectors for groups i and j , α is the marginal standard deviation, and ρ is the length-scale parameter.

3 Gaussian Processes

3.1 Posterior prediction under gaussian process prior

Notation. For convenience, we denote $g(\mathbb{E}[Y \mid X])$ as $g(\mu)$.

Theorem 1 (Posterior Prediction in Gaussian Process with Monotonic Link). *Assume a monotonic link function g such that:*

$$g(\mu) \sim \mathcal{N}(\boldsymbol{\mu}_{\text{prior},h}, \boldsymbol{\Sigma}_\mu[h, h'] + \mathbf{K}_\eta[k, k'] + \sigma^2 \mathbf{I}_n),$$

then the posterior prediction of f_* given $g(\mu)$ is:

$$f_* \mid g(\mu) \sim \mathcal{N}(\boldsymbol{\mu}_{f_*} + \boldsymbol{\Sigma}_{f_*y} \boldsymbol{\Sigma}_{yy}^{-1} (g(\mu) - \boldsymbol{\mu}_y), \boldsymbol{\Sigma}_{f_*f_*} - \boldsymbol{\Sigma}_{f_*y} \boldsymbol{\Sigma}_{yy}^{-1} \boldsymbol{\Sigma}_{yf_*}),$$

which we denote as:

$$f_* \mid g(\mu) \sim \mathcal{N}(\mathbb{E}[f_* \mid g(\mu)], \text{Cov}(f_* \mid g(\mu))),$$

and hence:

$$g(\mathbf{y}_*) \mid g(\mu) \sim \mathcal{N}(\mathbb{E}[f_* \mid g(\mu)], \text{Cov}(f_* \mid g(\mu)) + \sigma^2 \mathbf{I}_{n_*}).$$

Proof: The joint distribution of $\begin{bmatrix} g(\mu) \\ f_* \end{bmatrix}$ is:

$$\begin{bmatrix} g(\mu) \\ f_* \end{bmatrix} \sim \mathcal{N} \left(\begin{bmatrix} \boldsymbol{\mu}_{\text{prior},h} \\ \boldsymbol{\mu}_{\text{prior},h_*} \end{bmatrix}, \begin{bmatrix} \boldsymbol{\Sigma}_\mu[h, h'] + \mathbf{K}_\eta[k, k'] + \sigma^2 \mathbf{I}_n & \boldsymbol{\Sigma}_\mu[h, h_*] + \mathbf{K}_\eta[k, k_*] \\ \boldsymbol{\Sigma}_\mu[h_*, h] + \mathbf{K}_\eta[k_*, k] & \boldsymbol{\Sigma}_\mu[h_*, h_*] + \mathbf{K}_\eta[k_*, k_*] \end{bmatrix} \right),$$

which we denote as:

$$\begin{bmatrix} g(\mu) \\ f_* \end{bmatrix} \sim \mathcal{N} \left(\begin{bmatrix} \boldsymbol{\mu}_y \\ \boldsymbol{\mu}_{f_*} \end{bmatrix}, \begin{bmatrix} \boldsymbol{\Sigma}_{yy} & \boldsymbol{\Sigma}_{yf_*} \\ \boldsymbol{\Sigma}_{f_*y} & \boldsymbol{\Sigma}_{f_*f_*} \end{bmatrix} \right).$$

Applying the standard formula for conditional distributions of multivariate Gaussians, we obtain:

$$f_* \mid g(\mu) \sim \mathcal{N} \left(\boldsymbol{\mu}_{f_*} + \boldsymbol{\Sigma}_{f_*y} \boldsymbol{\Sigma}_{yy}^{-1} (g(\mu) - \boldsymbol{\mu}_y), \boldsymbol{\Sigma}_{f_*f_*} - \boldsymbol{\Sigma}_{f_*y} \boldsymbol{\Sigma}_{yy}^{-1} \boldsymbol{\Sigma}_{yf_*} \right).$$

Finally, because $g(\mathbf{y}_*) = f_* + \boldsymbol{\epsilon}_*$ with $\boldsymbol{\epsilon}_* \sim \mathcal{N}(0, \sigma^2 \mathbf{I}_{n_*})$, we conclude:

$$g(\mathbf{y}_*) \mid g(\mu) \sim \mathcal{N} \left(\mathbb{E}[f_* \mid g(\mu)], \text{Cov}(f_* \mid g(\mu)) + \sigma^2 \mathbf{I}_{n_*} \right).$$

3.2 Computational cost reduction

Theorem 2 (Computational Efficiency via Woodbury Identity). *The computational cost of inverting the $n \times n$ covariance matrix $\boldsymbol{\Sigma}_{\mathbf{y}\mathbf{y}}$ is reduced from $\mathcal{O}(n^3)$ to $\mathcal{O}(n^2(H + K))$, where H and K are the dimensions of the group-level covariates and $H, K \ll n$.*

Proof: Write the model as:

$$f + \boldsymbol{\epsilon} = Z_\mu \boldsymbol{\mu}_h + Z_\eta \boldsymbol{\eta}_k + \boldsymbol{\epsilon}$$

where $Z_\mu \in \mathbb{R}^{n \times H}$ and $Z_\eta \in \mathbb{R}^{n \times K}$ are group incidence matrices with entries

$$(Z_\mu)_{ih} = \mathbf{1}\{h[i] = h\}, \quad (Z_\eta)_{ik} = \mathbf{1}\{k[i] = k\},$$

and $\boldsymbol{\epsilon} \sim \mathcal{N}(0, \sigma^2 I_n)$. Then, the covariance matrix is:

$$V = Z_\mu \boldsymbol{\Sigma}_\mu Z_\mu^\top + Z_\eta K_\eta Z_\eta^\top + \sigma^2 I_n$$

Using the Woodbury identity:

$$V^{-1} = \left(\sigma^2 I_n + U C U^\top \right)^{-1} = \sigma^{-2} I_n - \sigma^{-2} U \left(C^{-1} + \sigma^{-2} U^\top U \right)^{-1} U^\top \sigma^{-2}$$

with $U = [Z_\mu \ Z_\eta] \in \mathbb{R}^{n \times (H+K)}$, $C = \text{diag}(\boldsymbol{\Sigma}_\mu, K_\eta)$.

The major costs are:

- $U^\top U$: $\mathcal{O}(n^2(H + K))$
- $(C^{-1} + U^\top U)^{-1}$: $\mathcal{O}((H + K)^3)$

Full expression: $\mathcal{O}(2n^2(H + K) + n(H + K)^2 + (H + K)^3)$, which simplifies to $\mathcal{O}(n^2(H + K))$.

Algorithm for Structured GP Posterior Inference (Full Poststratification Prediction)

Step 1: Construct Covariance Matrices

Compute $\Sigma_{yy}, \Sigma_{yf_*}, \Sigma_{f_*y}, \Sigma_{f_*f_*}$ using the estimated covariance matrices.

[The computational cost reduces from $\mathcal{O}(n^3)$ to $\mathcal{O}(n^2(H + K))$ and $\mathcal{O}(n_*^3)$ to $\mathcal{O}(n_*^2(H_* + K_*))$.]

Continuous outcome: $\Sigma_{yy} = Z_\mu \Sigma_\mu Z_\mu^\top + Z_\eta K_\eta Z_\eta^\top + \sigma^2 I_n$.

Binary outcome: $\Sigma_{yy} = Z_\mu \Sigma_\mu Z_\mu^\top + Z_\eta K_\eta Z_\eta^\top + \epsilon I_n$, $\epsilon > 0$ ensures positive definiteness.

Step 2: Cholesky Decomposition & Posterior Mean

Compute $L := \text{Cholesky}(\Sigma_{yy})$ and $g(\mu)_c := g(\mu) - \mu_y = f - \mu_y$.

Compute $V_1 := \text{solve}(L^\top, \text{solve}(L, g(\mu)_c)) = (LL^\top)^{-1}g(\mu)_c$

Then, $\mathbb{E}[f_* | g(\mu)] = \mu_{f_*} + \Sigma_{f_*y} V_1$

Step 3: Posterior Covariance

Compute $V_2 := \text{solve}(L, \Sigma_{yf_*})$

Then, $\text{Cov}(f_* | g(\mu)) = \Sigma_{f_*f_*} - V_2^\top V_2$

Step 4: Posterior Sampling

Identity link:

$$y_* | \mu \sim \mathcal{N}(\mathbb{E}[f_* | \mu], \text{Cov}(f_* | \mu) + \sigma^2 I_{n_*})$$

Binary outcome:

$$f_* | g(\mu) \sim \mathcal{N}(\mathbb{E}[f_* | g(\mu)], \text{Cov}(f_* | g(\mu)))$$

$$p_* \equiv g^{-1}(f_*)$$

$$y_* | f_* \sim \text{Bernoulli}(p_*)$$

For prediction at observed sample units, computation is restricted to the $K \times K$ Gaussian process kernel over the distinct poststratification cells present in the sample. Predictions are obtained via

Cholesky factorization of this cell-level kernel, avoiding construction of an $n \times n$ covariance matrix and reducing computational cost when $K \ll n$.

3.3 Connection to kernel ridge regression and crossed random effects model

Proposition 1 (Equivalence of KRR and GP Posterior Mean under Identity Link). *Consider our hierarchical Gaussian process model with an **identity link** $g(x) = x$, noise variance σ^2 , and suppose the group-level prior covariance $\Sigma_\mu \rightarrow 0$ (so that $\mu_h \equiv 0$). Let*

$$\mathbf{K} = K_\eta[k, k'],$$

the $n \times n$ kernel matrix over the n observed groups, and let $\mathbf{y} \in \mathbb{R}^n$ be the observed outcomes. Then the GP posterior mean at a new input x^ is*

$$\mathbb{E}[f(x^*) \mid \mathbf{y}] = \mathbf{k}(x^*, X)^\top (\mathbf{K} + \sigma^2 I_n)^{-1} \mathbf{y},$$

which coincides exactly with the Kernel Ridge Regression solution

$$\hat{f}(x^*) = \sum_{i=1}^n \alpha_i k(x_i, x^*) \quad \text{with} \quad \boldsymbol{\alpha} = (\mathbf{K} + \lambda I_n)^{-1} \mathbf{y} \quad \text{and} \quad \lambda = \sigma^2.$$

Proof: Under $\Sigma_\mu \rightarrow 0$, our latent function reduces to $f = \eta \sim \mathcal{GP}(0, K_\eta)$. With identity link and Gaussian noise,

$$\mathbf{y} \sim \mathcal{N}(0, \mathbf{K} + \sigma^2 I_n).$$

The standard GP posterior mean at x^* is

$$\mathbb{E}[f(x^*) \mid \mathbf{y}] = \mathbf{k}(x^*, X)^\top (\mathbf{K} + \sigma^2 I_n)^{-1} \mathbf{y},$$

where $\mathbf{k}(x^*, X)_i = k_\eta(x^*, x_i)$. Let $\mathcal{H}_k$ be a Reproducing Kernel Hilbert Space (RKHS) associated with a positive-definite kernel function $K(\cdot, \cdot)$. Then, KRR solves

$$\hat{f} = \arg \min_{f \in \mathcal{H}_k} \sum_{i=1}^n (y_i - f(x_i))^2 + \lambda \|f\|_{\mathcal{H}_k}^2$$

and admits the dual form

$$\hat{f}(x^*) = \mathbf{k}(x^*, X)^\top (\mathbf{K} + \lambda I_n)^{-1} \mathbf{y}.$$

Thus, setting $\lambda = \sigma^2$ makes the GP posterior mean and the KRR prediction identical.

Proposition 2. *Without loss of generality, let the number of covariates p be 3. Let $X \in \mathbb{R}^{K \times p}$ be the matrix of covariates for the K hierarchical groups, and set*

$$K_\eta = \tau^2 X X^\top.$$

Assume the identity link $g(x) = x$, observation noise $\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$, and a group-intercept prior $\mu \sim \mathcal{N}(\mu_{\text{prior}}, \Sigma_\mu)$. Let

$$b_0 \sim \mathcal{N}(\mu_{\text{prior}}, \Sigma_\mu) \quad b_1 \sim \mathcal{N}(0, \tau^2 I_p)$$

Then the hierarchical GP model

$$y = \mu_h + \eta_k + \varepsilon, \quad \eta \sim \mathcal{N}(0, K_\eta),$$

is equivalent to the crossed random effects model with a random intercept $b_{0,ij}$ and a random-slope $b_{1,ijk}$:

$$y_{ijk} = b_{0,ij} + X_{ijk} b_{1,ijk} + \varepsilon_{ijk}, \text{ where } i, j, k \text{ represent the levels of three covariates.}$$

Proof: By construction,

$$\eta \sim \mathcal{N}(0, K_\eta) = \mathcal{N}(0, \tau^2 X X^\top).$$

Introducing $\mathbf{u} \sim \mathcal{N}(\mathbf{0}, \tau^2 I_P)$ and setting $\eta = X \mathbf{u}$ yields

$$\text{Cov}(\eta) = X \text{Cov}(\mathbf{u}) X^\top = \tau^2 X X^\top = K_\eta, \quad \mathbb{E}[\eta] = \mathbf{0}.$$

Matching i, j to a group H and i, j, k to a group K gives

$$y_i = \mu_i + \eta_i + \varepsilon_i = b_{0,ij} + X_{ijk} b_{1,ijk} + \varepsilon_{ijk},$$

with the specified priors on b_0 and b_1 .

3.4 Uniform Convergence of the Posterior Predictive Mean

Theorem 3. *Consider the HGP regression model with additive structure*

$$f(x) = \mu_{h(x)} + \eta_{k(x)}, \quad y_i = f(x_i) + \varepsilon_i, \quad \varepsilon_i \stackrel{iid}{\sim} N(0, \sigma^2),$$

with priors $\mu \sim N(\mu_{\text{prior}}, \Sigma_\mu)$ and $\eta \sim GP(0, K_\eta)$, where K_η is a squared-exponential kernel over the K unique rows (cells) of X . Let $\Pi_n(\cdot) \equiv \Pi(\cdot \mid y_{1:n}, X_{1:n})$ denote the posterior over f , and define the posterior predictive mean

$$\hat{f}_n(x) \equiv \mathbb{E}_{\Pi_n}[f(x)].$$

Assume there exists a deterministic rate $\varepsilon_n \downarrow 0$ and constants $M > 0$, $c_0 > 0$ such that:

(C1) **Sup-norm contraction:**

$$\Pi_n\left(\|f - f_0\|_\infty > M\varepsilon_n\right) \xrightarrow{n \rightarrow \infty} 0 \quad \text{in } P_{f_0}\text{-probability.}$$

(C2) **Posterior second-moment boundedness:**

$$\sup_n \mathbb{E}_{f_0}[\mathbb{E}_{\Pi_n}\|f - f_0\|_\infty^2] < \infty$$

Then

$$\|\hat{f}_n - f_0\|_\infty = O_{P_{f_0}}(\varepsilon_n).$$

Proof: Fix the data $D_n = (y_{1:n}, X_{1:n})$. By Jensen's inequality and the triangle inequality,

$$\|\hat{f}_n - f_0\|_\infty = \sup_{x \in \mathcal{X}} \left| \mathbb{E}_{\Pi_n}[f(x)] - f_0(x) \right| \leq \mathbb{E}_{\Pi_n}\|f - f_0\|_\infty.$$

Let

$$A_n \equiv \{\|f - f_0\|_\infty \leq M\varepsilon_n\}.$$

Split the posterior expectation:

$$\mathbb{E}_{\Pi_n} \|f - f_0\|_\infty = \mathbb{E}_{\Pi_n} [\|f - f_0\|_\infty \mathbf{1}_{A_n}] + \mathbb{E}_{\Pi_n} [\|f - f_0\|_\infty \mathbf{1}_{A_n^c}].$$

The first term is bounded by $M\varepsilon_n$. For the second term, Cauchy–Schwarz gives

$$\mathbb{E}_{\Pi_n} [\|f - f_0\|_\infty \mathbf{1}_{A_n^c}] \leq \left(\mathbb{E}_{\Pi_n} \|f - f_0\|_\infty^2 \right)^{1/2} \Pi_n(A_n^c)^{1/2}.$$

By Markov's inequality and (C2), $(\mathbb{E}_{\Pi_n} \|f - f_0\|_\infty^2)^{1/2} = O_{P_{f_0}}(1)$. By (C1), $\Pi_n(A_n^c) \rightarrow 0$ in P_{f_0} -probability, hence $\Pi_n(A_n^c)^{1/2} = o_{P_{f_0}}(1)$. Therefore,

$$\mathbb{E}_{\Pi_n} [\|f - f_0\|_\infty \mathbf{1}_{A_n^c}] = o_{P_{f_0}}(1).$$

Combining the bounds,

$$\|\hat{f}_n - f_0\|_\infty \leq M\varepsilon_n + o_{P_{f_0}}(\varepsilon_n) = O_{P_{f_0}}(\varepsilon_n).$$

since $\varepsilon_n \downarrow 0$.

Corollary 1 (General monotone link g). *Assume the conditions of Theorem 3. In addition, suppose there exists an interval $\mathcal{I} \subset \mathbb{R}$ and a constant $L < \infty$ such that*

$$f_0(\mathcal{X}) \subset \mathcal{I}^\circ \quad \text{and} \quad \sup_{t \in \mathcal{I}} |(g^{-1})'(t)| \leq L.$$

Define

$$\mu_0(x) \equiv g^{-1}(f_0(x)), \quad \hat{\mu}_n(x) \equiv \mathbb{E}_{\Pi_n}[g^{-1}(f(x))].$$

Then

$$\|\hat{\mu}_n - \mu_0\|_\infty = O_{P_{f_0}}(\varepsilon_n).$$

Proof: Fix $x \in \mathcal{X}$. By Jensen,

$$|\hat{\mu}_n(x) - \mu_0(x)| = \left| \mathbb{E}_{\Pi_n}[g^{-1}(f(x))] - g^{-1}(f_0(x)) \right| \leq \mathbb{E}_{\Pi_n} \left| g^{-1}(f(x)) - g^{-1}(f_0(x)) \right|.$$

Define the events

$$A_n \equiv \{\|f - f_0\|_\infty \leq M\varepsilon_n\}, \quad B_n \equiv \{f(\mathcal{X}) \subset \mathcal{I}\}.$$

Since $f_0(\mathcal{X}) \subset \mathcal{I}^\circ$, there exists $\delta > 0$ such that $\{t : \text{dist}(t, f_0(\mathcal{X})) \leq \delta\} \subset \mathcal{I}$. Hence, for all sufficiently large n (with $M\varepsilon_n \leq \delta$),

$$A_n \subset B_n, \quad \text{equivalently} \quad B_n^c \subset A_n^c.$$

Therefore,

$$\Pi_n(B_n^c) \leq \Pi_n(A_n^c) = \Pi_n(\|f - f_0\|_\infty > M\varepsilon_n) \xrightarrow{P_{f_0}} 0$$

by the contraction assumption (C1) in Theorem 3, and consequently $\Pi_n(B_n) \xrightarrow{P_{f_0}} 1$.

On the event B_n , the mean value theorem and the derivative bound imply

$$\sup_{x \in \mathcal{X}} \left| g^{-1}(f(x)) - g^{-1}(f_0(x)) \right| \leq L \|f - f_0\|_\infty.$$

Therefore,

$$\|\hat{\mu}_n - \mu_0\|_\infty \leq L \mathbb{E}_{\Pi_n} \|f - f_0\|_\infty + o_{P_{f_0}}(1),$$

where the $o_{P_{f_0}}(1)$ term absorbs the contribution from B_n^c (since $\Pi_n(B_n^c) \rightarrow 0$ in P_{f_0} -probability).

By Theorem 3, $\mathbb{E}_{\Pi_n} \|f - f_0\|_\infty = O_{P_{f_0}}(\varepsilon_n)$, hence

$$\|\hat{\mu}_n - \mu_0\|_\infty = O_{P_{f_0}}(\varepsilon_n).$$

3.5 Connection to Survey Weighting

We now present a theorem that connects the regression function $g^{-1}(f)$ to population-calibrated survey weights.

Lemma 1 (Inverse-link linearization remainder). *Let $\phi \equiv g^{-1} : \mathbb{R} \rightarrow \mathbb{R}$ be the inverse link. Assume there exists an interval $I \subset \mathbb{R}$ such that $f_0(\mathcal{X}) \subset I^\circ$, ϕ is continuously differentiable on I , and $\sup_{t \in I} |\phi'(t)| < \infty$. Let $\hat{f}_n$ satisfy the conclusion of Theorem 3:*

$$\|\hat{f}_n - f_0\|_\infty = O_{P_{f_0}}(\varepsilon_n), \quad \varepsilon_n \rightarrow 0.$$

For design points $x_1, \dots, x_n \in \mathcal{X}$ define

$$z_i := f_0(x_i), \quad \hat{z}_i := \hat{f}_n(x_i), \quad z := (z_1, \dots, z_n)^\top, \quad \hat{z} := (\hat{z}_1, \dots, \hat{z}_n)^\top,$$

and let $A(\hat{z}) := \text{diag}(\phi'(\hat{z}_1), \dots, \phi'(\hat{z}_n))$. Then, on a neighborhood of the truth,

$$\phi(z) - \phi(\hat{z}) = A(\hat{z})(z - \hat{z}) + r_g,$$

where the remainder satisfies

$$\|r_g\|_\infty = o_{P_{f_0}}(\|z - \hat{z}\|_\infty) \quad \text{and hence} \quad \|r_g\|_\infty = o_{P_{f_0}}(1).$$

Proof. First note that

$$\|z - \hat{z}\|_\infty = \max_{1 \leq i \leq n} |\hat{f}_n(x_i) - f_0(x_i)| \leq \|\hat{f}_n - f_0\|_\infty = O_{P_{f_0}}(\varepsilon_n) = o_{P_{f_0}}(1).$$

Fix i . By the mean value theorem there exists ξ_i between z_i and $\hat{z}_i$ such that

$$\phi(z_i) - \phi(\hat{z}_i) = \phi'(\xi_i)(z_i - \hat{z}_i).$$

Define the remainder componentwise by

$$r_{g,i} := (\phi'(\xi_i) - \phi'(\hat{z}_i))(z_i - \hat{z}_i), \quad r_g := (r_{g,1}, \dots, r_{g,n})^\top,$$

so that $\phi(z) - \phi(\hat{z}) = A(\hat{z})(z - \hat{z}) + r_g$. Then

$$\frac{\|r_g\|_\infty}{\|z - \hat{z}\|_\infty} \leq \max_{1 \leq i \leq n} |\phi'(\xi_i) - \phi'(\hat{z}_i)|.$$

Fix $\eta > 0$. Since ϕ' is continuous on I , it is uniformly continuous on every compact $I_0 \subset I$. Because $f_0(\mathcal{X}) \subset I^\circ$ and $\|\hat{z} - z\|_\infty = o_{P_{f_0}}(1)$, there exists a compact $I_0 \subset I^\circ$ such that

$$P_{f_0}(z_i, \hat{z}_i \in I_0 \ \forall i) \rightarrow 1.$$

Hence there exists $\delta > 0$ such that for all $u, v \in I_0$, $|u - v| < \delta$ implies $|\phi'(u) - \phi'(v)| < \eta$.

By the mean value theorem, $|\xi_i - \hat{z}_i| \leq |z_i - \hat{z}_i| \leq \|z - \hat{z}\|_\infty$. Therefore,

$$\begin{aligned} P_{f_0} \left(\max_i |\phi'(\xi_i) - \phi'(\hat{z}_i)| > \eta \right) &\leq P_{f_0}(\|z - \hat{z}\|_\infty \geq \delta) + P_{f_0}(\exists i : \{z_i, \hat{z}_i\} \not\subset I_0) \\ &\rightarrow 0, \end{aligned}$$

since $\|z - \hat{z}\|_\infty = o_{P_{f_0}}(1)$.

Consequently,

$$\max_i |\phi'(\xi_i) - \phi'(\hat{z}_i)| \xrightarrow{P_{f_0}} 0,$$

which implies

$$\|r_g\|_\infty \leq \|z - \hat{z}\|_\infty \max_i |\phi'(\xi_i) - \phi'(\hat{z}_i)| = o_{P_{f_0}}(\|z - \hat{z}\|_\infty).$$

Since $\|z - \hat{z}\|_\infty = o_{P_{f_0}}(1)$, we conclude that $\|r_g\|_\infty = o_{P_{f_0}}(1)$.

□

Theorem 4 (Local equivalent unit weights). *Let*

$$z \equiv g(\mu) = g(\mathbb{E}[Y \mid X]) \in \mathbb{R}^n$$

denote the latent mean-scale vector, and let

$$f_* \in \mathbb{R}^K$$

denote the latent GP values at the K poststratification cells X_ . Under the setup of Theorem 3 and Corollary 1, the posterior predictive mean*

$$m_*(z) \equiv \mathbb{E}[f_* \mid z] = \mu_{f_*} + \Sigma_{f_*, y} \Sigma_{yy}^{-1} (z - \mu_y)$$

is affine in z .

Let N_k^{pop} be the population cell counts, $N = \sum_{k=1}^K N_k^{\text{pop}}$, and

$$N^{\text{pop}} = (N_1^{\text{pop}}, \dots, N_K^{\text{pop}})^\top \in \mathbb{R}^K.$$

Define the model-based poststratified estimand

$$T(z) \equiv \frac{1}{N} (N^{\text{pop}})^\top g^{-1}(m_*(z)),$$

where g^{-1} is applied componentwise. Fix a fitted value $\hat{z} \in \mathbb{R}^n$ satisfying

$$\|z - \hat{z}\|_\infty = o_{P_{f_0}}(1).$$

Define the local equivalent unit weights $w \in \mathbb{R}^n$ via the local survey-weight identity

$$T(z) - T(\hat{z}) = \frac{1}{n} w^\top \{g^{-1}(z) - g^{-1}(\hat{z})\} + o_{P_{f_0}}(\|z - \hat{z}\|_\infty). \quad (1)$$

Assume g^{-1} is continuously differentiable on a neighborhood of $m_*(\hat{z})$, and define

$$A(\hat{z}) = \text{diag}((g^{-1})'(\hat{z}_1), \dots, (g^{-1})'(\hat{z}_n)) \in \mathbb{R}^{n \times n},$$

$$D_* = \text{diag}((g^{-1})'(m_{*,1}(\hat{z})), \dots, (g^{-1})'(m_{*,K}(\hat{z}))) \in \mathbb{R}^{K \times K}.$$

Then w is uniquely determined (for large n) and satisfies

$$T(z) = T(\hat{z}) + \nabla T(\hat{z})^\top (z - \hat{z}) + o_{P_{f_0}}(\|z - \hat{z}\|_\infty), \quad w = n A(\hat{z})^{-1} \nabla T(\hat{z}),$$

with the closed-form expression

$$w = \frac{n}{N} A(\hat{z})^{-1} \Sigma_{yy}^{-1} \Sigma_{y,f_*} D_* N^{\text{pop}}.$$

Proof: From the proof of Lemma 1,

$$\|z - \hat{z}\|_\infty = o_{P_{f_0}}(1).$$

Step 1 (Linearization of the inverse link at the unit level). Lemma 1 yields

$$g^{-1}(z) - g^{-1}(\hat{z}) = A(\hat{z})(z - \hat{z}) + r_g, \quad \|r_g\|_\infty = o_{P_{f_0}}(\|z - \hat{z}\|_\infty).$$

Step 2 (Linearization at the poststratification-cell level). Writing $\phi \equiv g^{-1}$,

$$T(z) - T(\hat{z}) = \frac{1}{N} (N^{\text{pop}})^\top \{\phi(m_*(z)) - \phi(m_*(\hat{z}))\}.$$

Applying the argument of Lemma 1 componentwise to $m_*(z) \in \mathbb{R}^K$ gives

$$\phi(m_*(z)) - \phi(m_*(\hat{z})) = D_* \{m_*(z) - m_*(\hat{z})\} + r_{g,*}, \quad \|r_{g,*}\|_\infty = o_{P_{f_0}}(\|z - \hat{z}\|_\infty).$$

Since

$$m_*(z) - m_*(\hat{z}) = \Sigma_{f_*,y} \Sigma_{yy}^{-1} (z - \hat{z}),$$

we obtain

$$T(z) - T(\hat{z}) = \frac{1}{N} (N^{\text{pop}})^\top D_* \Sigma_{f_*,y} \Sigma_{yy}^{-1} (z - \hat{z}) + r_T, \quad r_T = \frac{1}{N} (N^{\text{pop}})^\top r_{g,*}.$$

Moreover,

$$|r_T| \leq \frac{1}{N} \|N^{\text{pop}}\|_1 \|r_{g,*}\|_\infty = o_{P_{f_0}}(\|z - \hat{z}\|_\infty).$$

Step 3 (Solving for the weights). Combining the preceding displays with (1) and matching the leading linear terms in $(z - \hat{z})$ yields

$$\frac{1}{n} w^\top A(\hat{z})(z - \hat{z}) = \nabla T(\hat{z})^\top (z - \hat{z}),$$

and hence

$$w = n A(\hat{z})^{-1} \nabla T(\hat{z}).$$

Finally, applying the chain rule gives

$$\nabla T(\hat{z}) = \frac{1}{N} \Sigma_{yy}^{-1} \Sigma_{y,f_*} D_* N^{\text{pop}},$$

which completes the proof. □

Special cases. Write

$$\eta_* \equiv m_*(\hat{z}) \in \mathbb{R}^K, \quad D_* = \text{diag}((g^{-1})'(\eta_{*,1}), \dots, (g^{-1})'(\eta_{*,K})),$$

and recall the unit-level inverse-link Jacobian

$$A(\hat{z}) = \text{diag}((g^{-1})'(\hat{z}_1), \dots, (g^{-1})'(\hat{z}_n)).$$

Then Theorem 4 yields the local equivalent unit weights

$$w = \frac{n}{N} A(\hat{z})^{-1} \Sigma_{yy}^{-1} \Sigma_{yf_*} D_* N^{\text{pop}}.$$

- **Identity (linear).** If $g^{-1}(\eta) = \eta$, then $(g^{-1})'(\eta) = 1$ for all η , so

$$A(\hat{z}) = I_n, \quad D_* = I_K.$$

Hence the weights reduce to

$$w_B = \frac{n}{N} \Sigma_{yy}^{-1} \Sigma_{yf_*} N^{\text{pop}},$$

which coincides with the standard linear-model local equivalent unit weights.

- **Logit.** If

$$g^{-1}(\eta) = \text{expit}(\eta) = \{1 + \exp(-\eta)\}^{-1},$$

define the fitted probabilities

$$p \equiv \text{expit}(\hat{z}) \in \mathbb{R}^n, \quad p_* \equiv \text{expit}(\eta_*) \in \mathbb{R}^K.$$

Since $(g^{-1})'(\eta) = \text{expit}(\eta)\{1 - \text{expit}(\eta)\}$, we have

$$A(\hat{z}) = \text{diag}(p_1(1 - p_1), \dots, p_n(1 - p_n)),$$

$$D_* = \text{diag}(p_{*,1}(1 - p_{*,1}), \dots, p_{*,K}(1 - p_{*,K})),$$

and therefore

$$w_B = \frac{n}{N} \text{diag}(p \odot (1 - p))^{-1} \Sigma_{yy}^{-1} \Sigma_{yf*} \text{diag}(p_* \odot (1 - p_*)) N^{\text{pop}},$$

where $\odot$ denotes elementwise multiplication.

- **Probit.** If $g^{-1}(\eta) = \Phi(\eta)$, let φ denote the standard normal pdf. Then $(g^{-1})'(\eta) = \varphi(\eta)$, so

$$A(\hat{z}) = \text{diag}(\varphi(\hat{z}_1), \dots, \varphi(\hat{z}_n)), \quad D_* = \text{diag}(\varphi(\eta_{*,1}), \dots, \varphi(\eta_{*,K})),$$

and

$$w_B = \frac{n}{N} \text{diag}(\varphi(\hat{z}))^{-1} \Sigma_{yy}^{-1} \Sigma_{yf*} \text{diag}(\varphi(\eta_*)) N^{\text{pop}}.$$

The uniform convergence result in Theorem 3 is closely related to classical posterior contraction results for Gaussian process priors (van der Vaart and van Zanten, 2008), which establish global convergence of the posterior distribution toward the true latent function under suitable regularity conditions. While those results are typically concerned with estimation accuracy of the latent function itself, the role of Theorem 3 in the present work is different. Here, uniform convergence of the posterior predictive mean serves as a regularity condition that enables differentiation of a population-level poststratified functional.

In particular, Theorem 3 guarantees that the fitted latent surface lies, with probability tending to one, in a uniform neighborhood of the truth on which the inverse-link expansion in Lemma 1 holds uniformly over the poststratification space. This uniform control ensures that the gradient of the poststratified estimand is well defined and stable in a neighborhood of the truth. It is this property that underlies the local linearization in Theorem 4, which characterizes survey weights as directional derivatives of the population estimand evaluated along the fitted latent surface. Thus, whereas classical contraction results justify pointwise or global estimation of latent functions, Theorem 3 provides the theoretical bridge needed to translate nonparametric GP regularity into a valid, differentiability-based representation of survey weights in high-dimensional and sparse poststratification settings.

4 Results

4.1 Simulation setup

We consider a synthetic population constructed from a full cross-classification of city ($C = 10$), age ($A = 6$), ethnicity ($E = 4$), education ($D = 5$), and marital status ($M = 2$), yielding $K = 2,400$ poststratification cells in total. The population size is set to $N_{\text{pop}} = 100,000$. From this population, we draw a simple random sample of size $n = 2,000$.

Due to sparsity induced by the high-dimensional cell structure, only a subset of cells are observed in the sample. In our simulation run, only 657 of the 2,400 poststratification cells are observed in the sample. Of these observed cells, population-level prior information on the mean outcome is available for 239 cells, highlighting the substantial sparsity and partial population coverage characteristic of the design. Model fitting is therefore based on the observed cells, with unobserved cells handled implicitly through the hierarchical prior structure. This reflects realistic survey settings in which many poststratification cells are unobserved or weakly observed.

All simulation settings are held fixed across scenarios, with variation arising only from the data-generating mechanism governing the true cell-level outcome probabilities.

4.2 Simulation Results

We compare three approaches: hierarchical Gaussian process weighting (HGP), logistic regression without priors (GLM), and logistic regression estimated by maximum a posteriori inference under group-level Gaussian priors (Logit MAP). Two data-generating mechanisms are considered: a correctly specified parametric logistic model and a nonparametric mechanism with latent Gaussian process structure across poststratification cells.

Table 1 reports results under two complementary evaluation regimes. Panel A presents oracle evaluation, in which cell-level mean squared error (MSE) is computed against the true population distribution. Panel B presents method-induced evaluation, in which the same squared errors are weighted using each method’s own estimated survey weights. Specifically, letting p_k denote the true outcome probability in poststratification cell k and $\hat{p}_k$ its estimate, we define

$$\text{MSE}_{\text{oracle}} = \frac{\sum_k N_k (\hat{p}_k - p_k)^2}{\sum_k N_k} \quad \text{and} \quad \text{MSE}_{\text{induced}} = \frac{\sum_k \hat{w}_k (\hat{p}_k - p_k)^2}{\sum_k \hat{w}_k},$$

where N_k denotes the true population count in cell k , $\hat{w}_i$ denotes the estimated survey weight for sampled unit i , and $\hat{w}_k$ denotes the method-induced weight assigned to cell k (constant within cell in our construction). We also report the unweighted MSE, obtained by averaging $(\hat{p}_k - p_k)^2$ equally across cells. The oracle criterion evaluates predictive accuracy under the true population distribution, while the method-induced criterion reflects downstream performance when the constructed survey weights implied by each fitted model are applied in practice.

Table 1 summarizes predictive performance under both oracle and method-induced evaluation regimes. Under the parametric data-generating mechanism, logistic regression without priors achieves the lowest mean squared error across all criteria, reflecting correct model specification. The hierarchical GP performs comparably but does not improve upon the correctly specified parametric model, while the MAP estimator incurs higher error due to regularization.

Table 1: Simulation results under parametric and nonparametric data-generating mechanisms. Panel A reports oracle evaluation using the true population cell counts N_k . Panel B reports method-induced evaluation using method-specific estimated weights (diagonal entries).

Panel A: Oracle evaluation (true cell counts N_k)			
Scenario	Method	Unweighted MSE	Population-weighted MSE
Parametric	HGP	0.0036	0.0030
	GLM (no prior)	0.0022	0.0022
	Logit (MAP)	0.0086	0.0084
Nonparametric (GP)	HGP	0.043	0.035
	GLM (no prior)	0.049	0.049
	Logit (MAP)	0.056	0.050

Panel B: Method-induced evaluation (closed-form estimated weights $\hat{w}_k$)		
Scenario	Method	MSE using own estimated weights
Parametric	HGP	0.0025
	GLM (no prior)	0.0021
	Logit (MAP)	0.0081
Nonparametric (GP)	HGP	0.028
	GLM (no prior)	0.044
	Logit (MAP)	0.033

In contrast, under the nonparametric (GP) data-generating mechanism, the hierarchical GP attains the lowest error under both oracle and method-induced evaluation. The performance gap is most pronounced under population-weighted oracle MSE, indicating that the GP effectively

captures latent structure in population-dominant cells that is not recoverable by parametric alternatives. Method-induced evaluation further highlights the importance of coherence between prediction and weighting. Panel B evaluates squared error using method-specific estimated weights, reflecting downstream performance when the constructed survey weights are applied in practice. Across all methods, method-induced evaluation yields lower MSE than oracle weighting, with the largest reductions occurring when predictions and weights are generated by the same model.

For the hierarchical GP, this coherence is particularly pronounced. The resulting weights inherit smooth, cell-level structure from the latent Gaussian process prior, allowing mass to be reallocated toward regions of the covariate space with higher population density while still borrowing strength across sparse cells. This alignment between prediction and weighting explains the substantial gains observed under method-induced evaluation in the nonparametric setting.

The comparison also clarifies the role of prior regularization in the logistic MAP estimator through the lens of equivalent weights. As shown in prior work, closed-form logistic equivalent weights arise from linearizing the estimating equations and depend only on auxiliary variables under correct specification. In the MAP setting, this construction is modified through the posterior mode and an information matrix augmented by prior precision, yielding stabilized but not fully calibrated weights.

Under the nonparametric data-generating mechanism, this stabilization reduces variance relative to the no-prior GLM but does not fully recover the population-level weighting implied by the latent structure. As a result, MAP improves robustness in sparse regions yet remains less well aligned with the oracle benchmark than the hierarchical GP, highlighting a bias–variance tradeoff relative to the more flexible nonparametric approach.

Overall, the strong alignment between oracle and method-induced evaluation for the hierarchical GP indicates that the proposed weighting procedure is not merely numerically convenient but substantively meaningful. Although the weights are not explicitly calibrated to population cell counts, they behave approximately as locally equivalent to population weighting for the mean functional, in the sense of Theorem 4. This provides empirical support for using the proposed weights in downstream survey estimation when exact calibration is either infeasible or undesirable.

4.3 Visual diagnostics by H -groups and K -cells

To complement the MSE summaries in Table 1, we visualize prediction accuracy at two aggregation levels. Recall that K indexes the full poststratification cells (city $\times$ age $\times$ ethnicity $\times$ education $\times$ marital status), while H indexes a coarser partition (city $\times$ age $\times$ ethnicity) that is used for the hierarchical prior on group-specific intercept offsets.

To complement the MSE summaries in Table 1, we present visual diagnostics at two levels of aggregation: H -groups and K -cells. In both cases, the diagnostics focus on the most informative portions of the sample by restricting attention to the top 50 groups ranked by sample frequency. This choice emphasizes regions of the covariate space that are most directly supported by the observed data, while still allowing population structure to influence the displayed rates.

For the H -level diagnostic (Figure 1), we first select the top 50 H -groups ranked by sample size. Each H -group consists of multiple finer poststratification cells (K -cells). Population and predicted rates are computed by aggregating over these constituent K -cells using population weights N_k :

$$p_H = \frac{\sum_{k \in H} N_k p_k}{\sum_{k \in H} N_k}, \quad \hat{p}_H = \frac{\sum_{k \in H} N_k \hat{p}_k}{\sum_{k \in H} N_k}.$$

This weighting is essential because K -cells within an H -group can differ substantially in population size. Population-weighted aggregation ensures that the H -level diagnostic reflects agreement under the true population distribution that drives downstream estimands, such as population means or totals obtained after applying the survey weights, rather than treating all constituent cells equally.

For the K -level diagnostic (Figure 2), we instead examine the top 50 K -cells ranked by sample size and compare predicted and population rates directly at the cell level. Since each K -cell is already the atomic unit of poststratification, there is no within-cell aggregation step, and the diagnostic reports raw cell rates $\hat{p}_k$ and p_k . This view isolates cell-level behavior in well-observed regions of the sample and provides a complementary perspective to the population-weighted H -group summaries.

Predicted vs Population Rates (Top H groups by sample size)

Top row: Parametric scenario | Bottom row: GP scenario

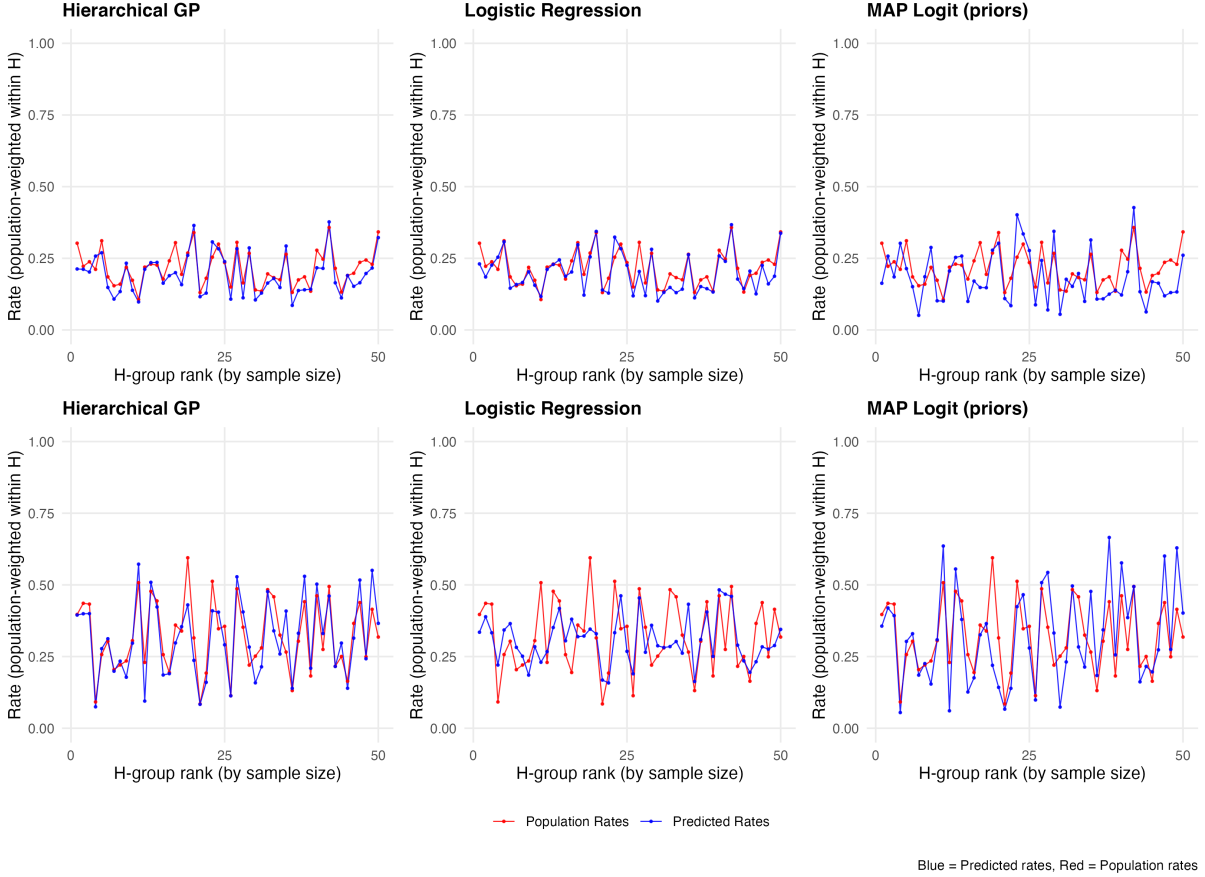


Figure 1: Predicted versus population rates for the top H -groups ranked by sample size. Within each H -group, both population and predicted rates are computed as population-weighted averages across constituent K -cells using N_k .

Figure 1 focuses on prediction accuracy at the H -group level, where both population and predicted rates are computed as population-weighted averages across constituent K -cells. In the parametric scenario, all three methods track the population rates reasonably well, reflecting the dominance of large- N_k cells under correct specification. Logistic regression without priors exhibits the closest alignment overall, consistent with its lowest population-weighted oracle MSE, while the hierarchical GP displays mild smoothing across some groups due to partial pooling. The MAP estimator applies global shrinkage to group-level effects, which stabilizes estimates in sparse settings but does not enforce smoothness across neighboring H -groups. As a result, predicted rates can still exhibit pronounced fluctuations across groups, even at this coarser level of aggregation.

Under the GP scenario, clearer differences emerge across methods. The hierarchical GP tracks

the population-weighted H -level rates more faithfully across groups, preserving both the overall level and the relative ordering of peaks and troughs. In contrast, the parametric baselines exhibit systematic discrepancies in several H -groups, reflecting their inability to adapt to latent nonparametric structure that varies smoothly across cells. These patterns are especially visible in groups where the population mass is spread across heterogeneous K -cells, highlighting the advantage of the GP-based borrowing of strength even after population-weighted aggregation.

Top-50 K -groups (by sample size): Predicted vs Population cell rates

Top row: Parametric scenario | Bottom row: GP scenario

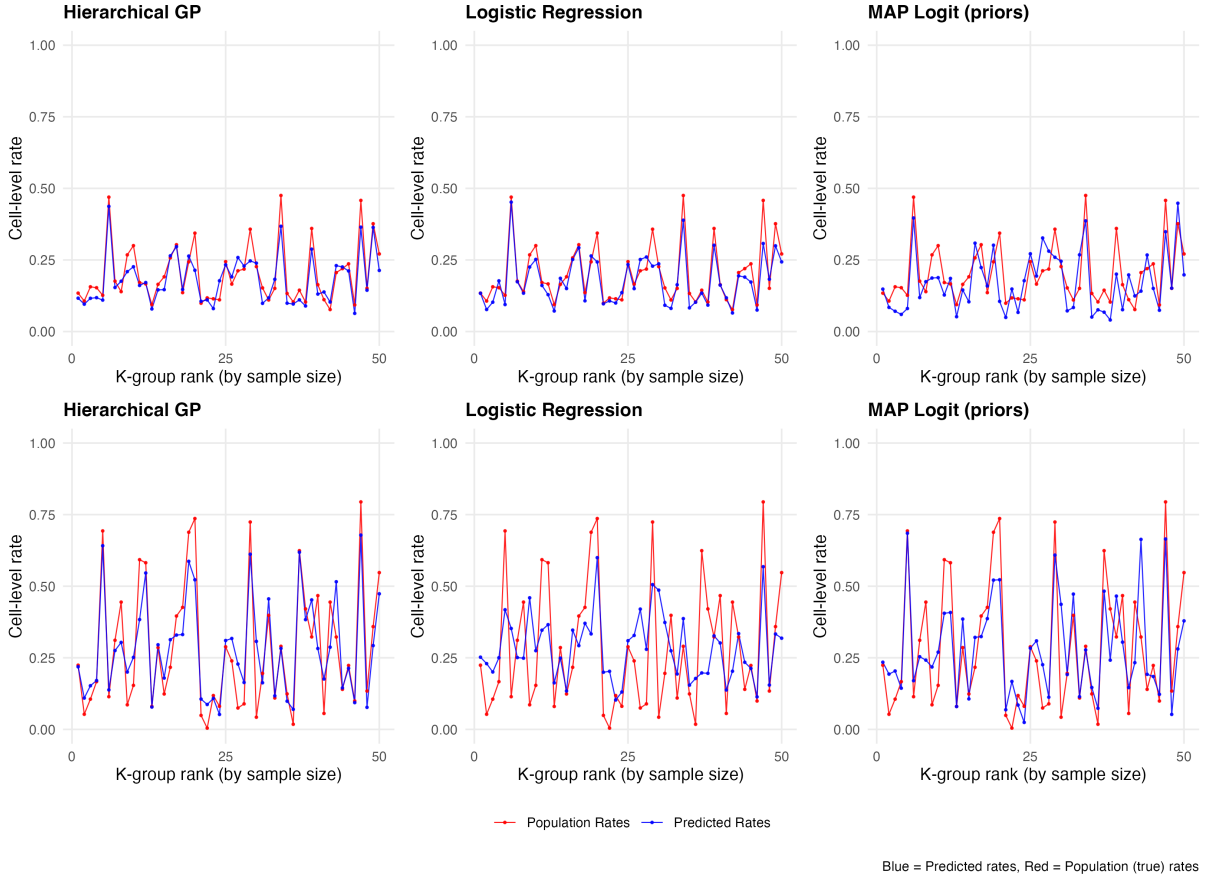


Figure 2: Predicted versus population *cell* rates for the top K -cells ranked by sample size. Each panel compares $\hat{p}_k$ (blue) and p_k (red) at the cell level, without additional within-cell aggregation.

Figure 2 presents diagnostics at the K -cell level, comparing predicted and population cell rates for the top K -cells ranked by sample size, without additional within-cell aggregation. This view isolates cell-level behavior in well-observed regions of the sample and provides a complementary perspective to the population-weighted H -group summaries.

In the parametric scenario, logistic regression without priors closely tracks the population cell

rates, consistent with correct specification at the individual cell level. The hierarchical GP remains competitive but exhibits mild smoothing across neighboring cells due to partial pooling. The MAP estimator produces stable predictions in sparse cells through shrinkage toward group-level effects, which can reduce extreme noise but may also distort fine-scale variation.

Under the GP scenario, sharper distinctions emerge across methods. The hierarchical GP more reliably reproduces the cell-to-cell variation in the population rates, capturing both large excursions and local irregularities among frequently observed cells. In contrast, the parametric baselines struggle to track this structure. GLM often fails to adapt to localized deviations, while Logit (MAP) captures heterogeneity in sparse cells through shrinkage but lacks the structured borrowing of strength needed to recover the underlying population patterns. This clarifies why population-weighted oracle MSE alone can be insufficient for assessing cell mean recovery: a method may perform well on high-mass cells while still misrepresenting the fine-grained structure revealed at the K -cell level.

4.4 Population mean calibration and local equivalent weights

The mean squared error criteria in Table 1, together with the cell-level diagnostics in Figures 1 and 2, assess predictive accuracy under the population distribution. As a complementary diagnostic, we directly evaluate how well each weighting scheme preserves the population mean functional, which is the primary object of interest in many survey applications and the target of the local equivalent weighting result in Theorem 4.

Let

$$\bar{p}^{\text{pop}} = \frac{\sum_k N_k p_k}{\sum_k N_k} \quad \text{and} \quad \bar{p}^w = \frac{\sum_k \hat{w}_k \hat{p}_k}{\sum_k \hat{w}_k},$$

denote the true population mean and the model-based weighted mean implied by the estimated survey weights, respectively.

We summarize mean-level agreement using the mean gap

$$\Delta_{\text{mean}} = \bar{p}^{\text{pop}} - \bar{p}^w,$$

and the relative gap

$$\Delta_{\text{rel}} = \left| \frac{\Delta_{\text{mean}}}{\bar{p}^{\text{pop}}} \right|.$$

Table 2: Population mean agreement under parametric and nonparametric data-generating mechanisms. Reported quantities compare the true population mean $\bar{p}^{\text{pop}}$ with the model-implied weighted mean $\bar{p}^w$.

Scenario	Method	$\bar{p}^{\text{pop}}$	$\bar{p}^w$	Relative gap (%)
Parametric	HGP	0.226	0.201	11.1
	GLM (no prior)	0.226	0.221	2.5
	Logit (MAP)	0.226	0.212	6.3
Nonparametric (GP)	HGP	0.323	0.333	3.0
	GLM (no prior)	0.323	0.319	1.2
	Logit (MAP)	0.323	0.314	3.0

Table 2 reports population mean agreement across methods. Under both data-generating mechanisms, all three approaches achieve small relative discrepancies, with mean gaps on the order of a few percentage points. This behavior is consistent with the local equivalence guarantees of Theorem 4, especially in the nonparametric setting where exact global calibration is neither imposed nor expected. Logistic regression without priors yields the smallest mean gap, reflecting its strong calibration of marginal totals under correct specification. The hierarchical GP and MAP estimators exhibit slightly larger global discrepancies but remain within single-digit relative error under the nonparametric mechanism.

These differences at the level of a single scalar functional should be interpreted with caution. As demonstrated by the subgroup-level diagnostics, the hierarchical GP achieves substantially improved local prediction accuracy across heterogeneous cells, even in settings where its global mean discrepancy is marginally larger than that of the unregularized logistic model. This behavior is consistent with the theoretical role of the GP-based weights, which prioritize smoothness and stability across the poststratification space rather than exact global calibration of a single moment.

Taken together, the MSE summaries, the visual comparisons, and population mean diagnostics confirm that the proposed HGP-based weights achieve local equivalence to population weighting for the mean functional while substantially improving accuracy under nonparametric structure. In high-dimensional and sparse survey settings, these results highlight the importance of evaluating

weighting methods using both global and local criteria, rather than relying solely on agreement at a single population-level summary.

4.5 Application to FFCWS Data

We apply the proposed hierarchical Gaussian process (HGP) weighting framework to data from the Fragile Families and Child Wellbeing Study (FFCWS), a longitudinal birth cohort study of children born in large U.S. cities between 1998 and 2000 (Reichman et al, 2001). A detailed description of the survey design, sampling scheme, and construction of poststratification cells is provided in our earlier work (Lee and Gelman, 2026). Here we briefly summarize the key features relevant for the present analysis.

The FFCWS design induces a high-dimensional poststratification structure formed by cross-classifying maternal marital status, race/ethnicity, age, education, and city of birth (Carlson, 2008). In the present application, population information is available for 3,339 poststratification cells out of a total of 3,840 possible K -cells defined by this cross-classification. We restrict attention to the $n = 3,424$ sampled observations whose corresponding K -cells have known population counts, yielding coverage of 1,059 unique K -cells within the population-supported subset. As in the simulation study, this structure results in substantial sparsity, with many population cells weakly observed or entirely unobserved in the sample.

Consistent with the simulation study, we choose to rely on population-level information at a coarser level of aggregation rather than at the full K -cell resolution. In particular, reliable external benchmarks are incorporated for $H = 303$ coarser groups defined by race/ethnicity, age, and city, rather than for all 1,059 observed K -cells. This choice reflects standard practice in applied survey settings, where the outcome of interest may be misclassified or unavailable at fine levels of stratification, necessitating the use of auxiliary information from external sources at aggregated resolutions. This design choice aligns naturally with the hierarchical GP framework, which is specifically constructed to propagate information from well-supported coarse groups to sparser fine-scale cells through structured partial pooling.

For the outcome variable, we use the father’s interview status, which is fully observed in the analytic sample and therefore avoids additional complications due to item nonresponse. The post-stratification covariates consist exclusively of ordinal and nominal categorical variables: ordinal

variables are treated as numerical inputs, while nominal variables are encoded using one-hot binary indicators (Garrido-Merch´an and Hern´andez-Lobato, 2020). This representation facilitates construction of the squared exponential kernel used in the Gaussian process component. Together, the high dimensionality of the poststratification space, uneven sample support across cells, and partial population coverage make FFCWS a demanding environment for survey weighting. These features render it a natural empirical test case for hierarchical Gaussian process smoothing, which is designed to borrow strength across related cells while maintaining local adaptivity.

Predicted vs Population Rates Ordered by Sample Size

Top row: H groups (coarser aggregation) | Bottom row: K groups (poststratification cells)
Red = N_k -weighted population benchmark; Blue = model-implied predicted rates

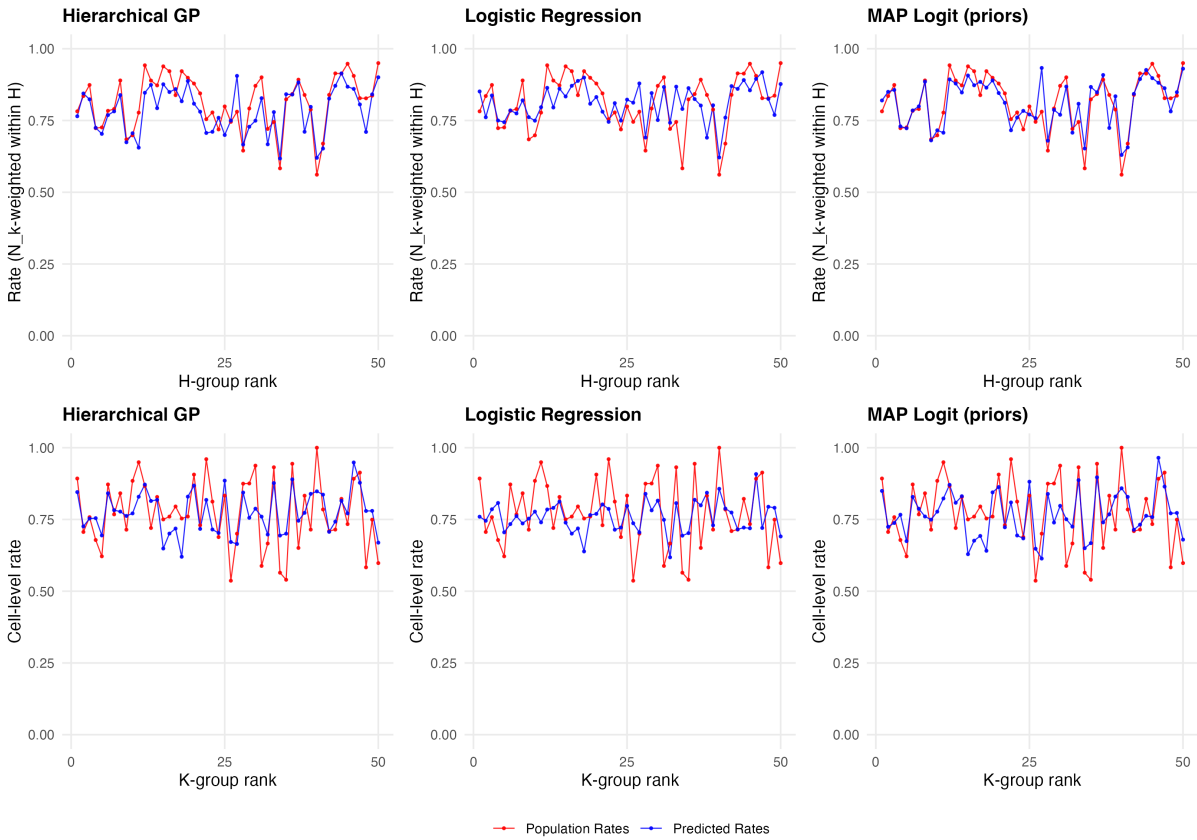


Figure 3: Predicted versus population rates ordered by sample size. The top row reports comparisons at the H -group level (coarser aggregation), while the bottom row reports comparisons at the K -cell level (poststratification cells). Within each panel, the model-implied predicted rates $\hat{p}$ (blue) are plotted against the corresponding N_k -weighted population benchmark rates p (red), with groups ranked by their sample frequencies. Results are shown for the hierarchical Gaussian process, logistic regression, and MAP logit models.

Figure 3 summarizes predictive performance in the FFCWS application by comparing model-

implied predicted rates with population benchmark rates ordered by sample size. The top row of Figure 3 reports comparisons at the H -group level, where both population and predicted rates are computed as N_k -weighted averages across constituent poststratification cells, while the bottom row reports comparisons at the K -cell level. In all panels, population benchmarks are constructed using available population counts, so that the diagnostics reflect predictive accuracy under the population distribution rather than equal weighting across cells.

Across both aggregation levels, clear differences emerge in how the methods track the population benchmarks. At the H -group level, agreement is strongest for the hierarchical GP and MAP logit, with predicted rates closely following the overall level and major fluctuations in the benchmark curves. This behavior is expected, since auxiliary population information and prior structure are imposed directly at the H -group resolution, so poststratified targets are most tightly disciplined at this level of aggregation. At the finer K -cell level, discrepancies across all methods become more pronounced, reflecting substantial sparsity and the absence of direct cell-level population constraints. In this regime, the hierarchical GP continues to provide coherent interpolation across related cells through structured partial pooling, while the unregularized logistic model exhibits more volatile cell-to-cell deviations and MAP logit stabilizes extremes via shrinkage without enforcing smoothness across neighboring cells. Overall, these patterns closely mirror the qualitative behavior observed in the simulation study, where alignment with population benchmarks was also strongest at the level of aggregation at which auxiliary information was supplied.

Table 3: Population mean agreement in the FFCWS application. Reported quantities compare the population benchmark mean $\bar{p}^{\text{pop}}$ (computed via poststratification of the observed sample outcome) with the model-implied weighted mean $\bar{p}^w$. The relative gap is reported in absolute value.

Method	$\bar{p}^{\text{pop}}$	$\bar{p}^w$	Relative gap (%)
HGP	0.793	0.811	2.2
GLM (no prior)	0.793	0.832	4.9
Logit (MAP)	0.793	0.811	2.2

Table 2 reports population mean agreement in the FFCWS application by comparing the benchmark population mean, obtained via poststratification of the observed sample outcome, with the model-implied weighted mean under each weighting scheme. All three methods achieve small relative gaps, remaining within a few percentage points of the benchmark, consistent with the

local equivalence guarantees of Theorem 4. However, unlike in the simulation study—where the unregularized logistic model achieved the smallest global mean discrepancy under correct specification—the hierarchical GP and MAP logit now attain the closest agreement with the population benchmark, while logistic regression without priors exhibits a noticeably larger gap. This reversal reflects the fact that, in the empirical setting, improved local predictions across heterogeneous subgroups can accumulate into better global accuracy when the population mass is unevenly distributed across cells.

Importantly, these results illustrate that agreement with a single global functional, such as the population mean, is not in itself a sufficient criterion for comparing weighting methods. In the FFCWS application, the hierarchical GP delivers superior local predictive performance across both H -groups and K -cells while simultaneously improving global mean accuracy relative to parametric alternatives. This underscores the value of evaluating survey weighting methods using a combination of local predictive diagnostics and global benchmarks, particularly in high-dimensional and sparse designs where calibration-based approaches may mask substantial local misalignment.

Table 4: Implied population counts by race/ethnicity under different weighting schemes. Entries report the total population mass allocated to each group, computed as $\sum_i w_i (N/n)$. Race/ethnicity categories follow the survey coding and are used solely for diagnostic comparison across weighting methods.

Method	White	Black	Hispanic	Other
Population (design-based)	430k	259k	353k	88k
Hierarchical GP	395k	341k	309k	86k
GLM (no prior)	440k	331k	293k	67k
Logit (MAP)	314k	399k	332k	85k

Table 4 reports the implied population mass allocated to race/ethnicity groups under each weighting scheme, computed as $\sum_i w_i (N/n)$. Unlike the simulation study, where population cell counts are known by construction, this diagnostic is reported only for the empirical application to assess how different weighting approaches redistribute population mass across demographic groups. The hierarchical GP produces implied counts that remain broadly aligned with the design-based population totals while avoiding the more extreme reallocations observed under purely calibration-based weights. Although these quantities are not targets of formal calibration in our framework, their stability provides additional reassurance that the proposed weighting procedure yields rea-

sonable population representations across key subgroups. Taken together with the predictive and mean-level diagnostics, this result further supports the practical reliability of the hierarchical GP weights in applied survey settings.

5 Discussion

In our previous work, we demonstrated that logistic regression provides a principled model-based construction of survey weights, with stable finite-sample behavior and strong asymptotic guarantees. However, logistic weighting is inherently limited by its reliance on a globally linear predictor, which can perform poorly in the presence of sparse poststratification cells, high-order interactions, and smooth nonlinear variation across structured populations.

The hierarchical Gaussian process (HGP) framework generalizes this approach by treating unit-level equivalent weights not as ad hoc adjustments, but as derivatives of a smooth latent mean surface over the poststratification space. Through a local linearization of the population estimand, the resulting weights admit an influence-function interpretation, quantifying each unit’s marginal contribution to the population mean under smooth perturbations of the fitted surface. Because the latent mean surface is modeled using a hierarchical Gaussian process, these derivatives inherit smooth structure across poststratification cells, yielding locally adaptive weights rather than global parametric corrections.

An important implication of the uniform convergence result in Theorem 3 is that the stability of the proposed weights is not merely empirical. Because the posterior predictive mean converges uniformly to the true latent surface, the derivatives defining the local equivalent weights also stabilize uniformly over the poststratification space. This ensures that the increased flexibility of the HGP does not lead to uncontrolled or erratic weight behavior, even in high-dimensional and sparse settings, combining nonparametric adaptivity with theoretical guarantees typically associated with parametric approaches.

From a practical perspective, HGP-based weighting offers several advantages over traditional and regression-based methods. By allowing information to be shared smoothly across related groups, it eliminates the need for manual collapsing of poststratification cells and substantially reduces sensitivity to empty and low-count cells. The resulting weights remain interpretable through

their local equivalent form, preserving compatibility with classical survey workflows.

Importantly, these practical benefits are reflected in how the method performs under different evaluation criteria. The comparison between oracle and weighting-method-induced evaluation highlights an often overlooked point: weighting methods should be assessed not only by predictive accuracy under the true population distribution, but also by coherence between prediction and weighting. Because the proposed weights are derived directly from the fitted latent surface, the HGP exhibits strong alignment between estimation and weighting, which explains its superior performance under weighting-method-induced evaluation in nonparametric settings.

Empirically, HGP-based weighting introduces greater heterogeneity in the resulting weights, leading to a lower effective sample size relative to logistic regression. This behavior is expected and principled. Because the HGP can capture localized nonlinear structure that is necessarily smoothed away under a globally linear specification, it reallocates weight mass more selectively toward units that are most informative for recovering population structure. In this sense, weight concentration reflects increased resolution rather than instability, trading effective sample size for reduced bias and improved estimation accuracy. This behavior is especially apparent under weighting-method-induced evaluation, where coherence between prediction and weighting is critical for performance.

Regularization in the HGP operates at the level of the latent function through a combination of hierarchical decomposition informed by population-level priors and kernel-induced smoothing, yielding shrinkage that is locally adaptive across the poststratification space. In contrast, logistic regression estimated via maximum a posteriori (MAP) imposes global regularization through coefficient shrinkage toward prior means, providing a computationally efficient intermediate approach that recovers much of the large-scale structure captured by the HGP while remaining less locally adaptive. Crucially, the survey setting enables population-aware regularization through known poststratification cell counts and auxiliary information, allowing both the prior and the estimand to reflect population structure directly—an advantage unavailable in standard machine-learning applications.

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